

Geometric Structure in Neural Population Codes

Subspace geometry, causal interventions, and validity in Steinmetz 2019

Elliott Tower

2026

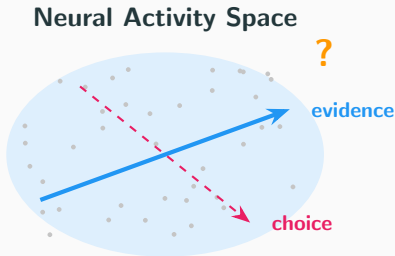
Why geometry?

Most neural analysis asks *what* neurons encode. We wanted to ask what **shape** that encoding takes—and whether the shape is causally meaningful, not just descriptive.

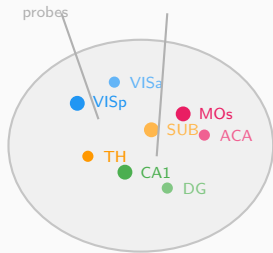
Geometric methods from ML interpretability (CKA, subspace analysis, interchange interventions) have rarely been applied to real neural data with proper validity checks.

Key question:

Do neural populations encode task variables in low-dimensional subspaces, and can we prove it causally?



The data: Steinmetz et al. 2019

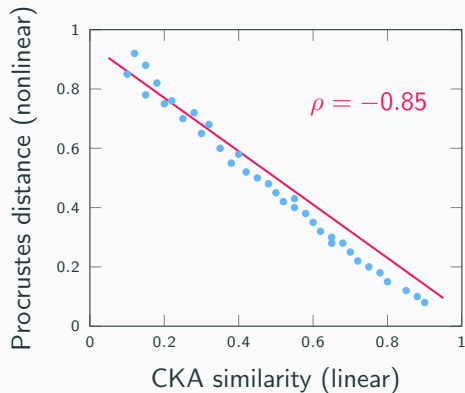


- 39 sessions, 10 mice
- **73 brain regions** recorded simultaneously
- Neuropixels probes (high density)
- Visual discrimination task (which side has higher contrast?)
- Trial variables: stimulus evidence, choice, reaction time, feedback

Steinmetz et al., Nature 2019

What geometry looks like

Linear and nonlinear geometry are anti-correlated



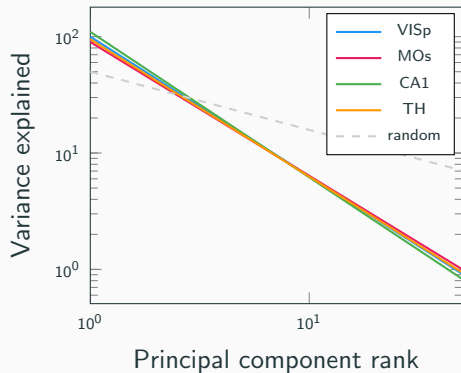
Regions that look similar in linear geometry look *different* in nonlinear geometry, and vice versa. This was not expected—most prior work treats these as interchangeable.

Implication:

You cannot characterize neural geometry with a single metric family. The choice of tool determines what you find.

Validated: Held 5 $\rightarrow$ 39 sessions; $p \approx 0$

Eigenvalue shapes are universal; trial content is not



Universal: eigenvalue spectra

Power-law decay, $r = 0.936$ across regions

Idiosyncratic: trial-by-trial modes

Same shape, different content ($r = 0.089$)

The scaffolding is shared but what gets built on it is unique to each region.

Neural choice manifolds are flat (no topological loops)

Expected?



$\beta_1 > 0$ (loop)

Found



$\beta_1 = 0$ (flat)

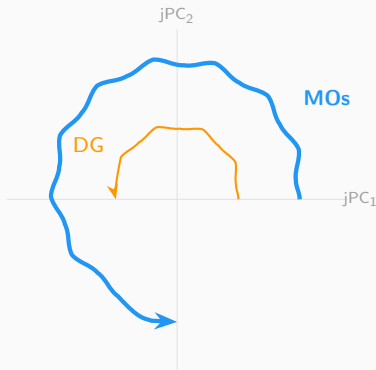
64/73 regions

Persistent homology shows choice representations are topologically trivial—no loops, no holes. For a binary choice task, the simplest possible topology turns out to be the actual topology.

This is a **negative result** but informative: rules out ring attractor models for this task.

Computed via Ripser (Vietoris-Rips filtration)

Every region rotates (but at different speeds)



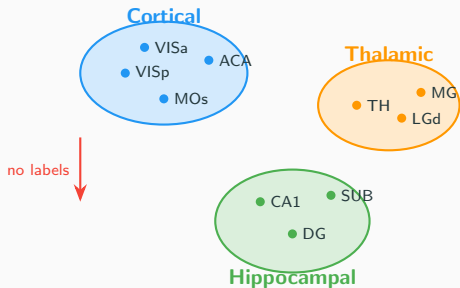
72/73 above shuffle null

Rotational dynamics (jPCA) appear in nearly every recorded region, but strength varies $10\times$. Motor and frontal regions rotate fastest; hippocampal regions rotate weakest.

$10\times$ variation in rotation strength
Frontal/motor $\gg$ sensory $>$ hippocampal

jPCA, Churchland et al. 2012

Geometry recovers anatomy (without anatomy labels)



Input: CKA similarity matrix (geometry only)

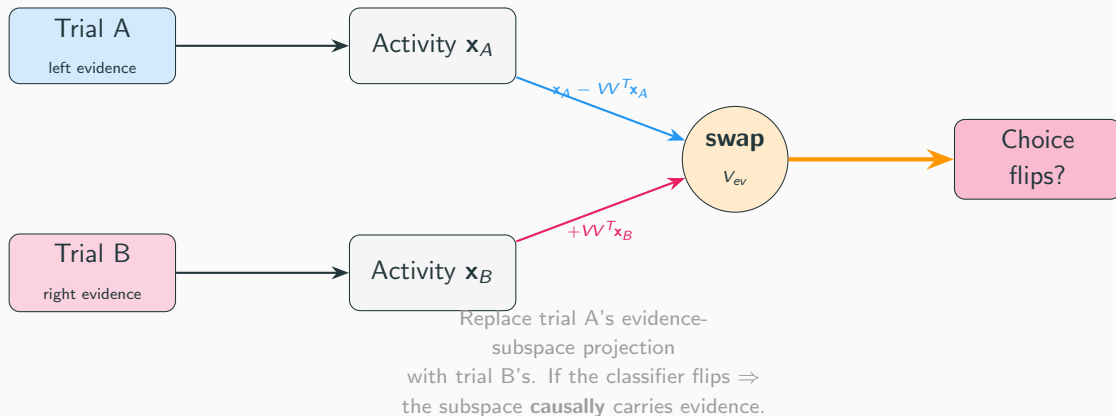
Output: $k = 3$ clusters

Result: matches cortical / thalamic / hippocampal split

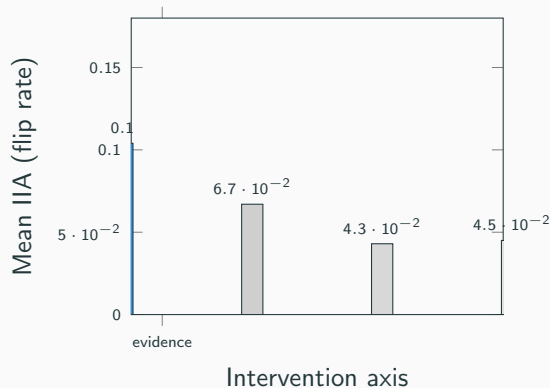
Geometry alone recovers the anatomical organization of the brain—no anatomy labels used as input. Brain regions with similar representational geometry are anatomically connected.

Causal interventions

Interchange Intervention Analysis (IIA)



IIA works: swapping evidence flips choices

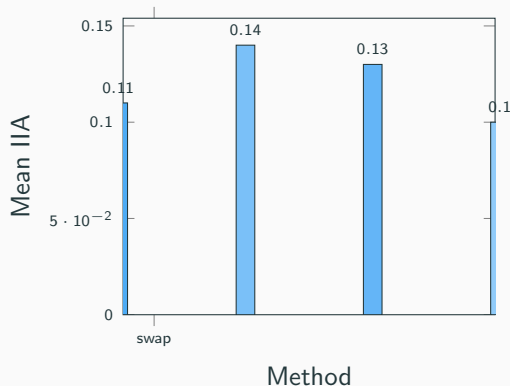


Specificity ratio: $2.3\times$ (evidence vs random)

The evidence subspace is not just correlated with choice—intervening on it *changes* the predicted choice. Swapping other axes (reaction time, feedback, random) does not produce the same effect. This is the key causal result: the evidence subspace has a specific causal role in carrying evidence to the choice computation.

From exp49 (specificity):
 $p < 0.001$ evidence vs random
(Mann-Whitney)

Four intervention methods agree



Cross-method rank correlation:

$\bar{\rho} = 0.72$ across all pairs

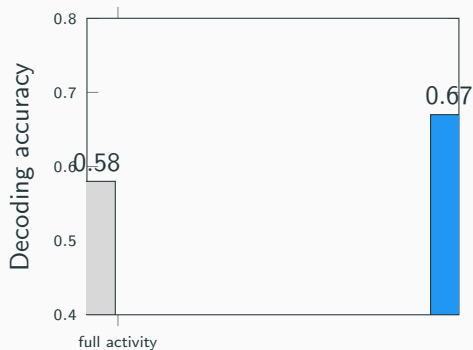
Regions that show high IIA with one method show high IIA with all methods.

The effect is not an artifact of the specific intervention method—method-invariance is the gold standard for causal claims.

Pass: $\rho \geq 0.7$ threshold (exp52)

Validation

The evidence subspace is sufficient for decoding



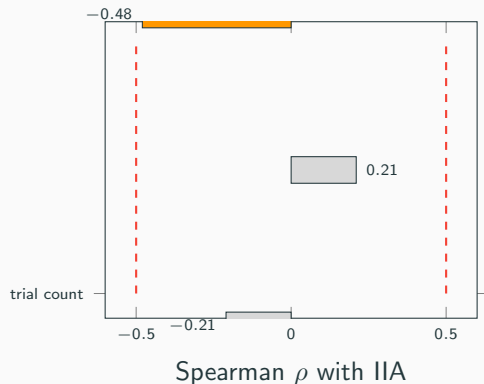
Recovery fraction = 1.16

The evidence subspace alone decodes choice *better* than full activity (acts as denoiser).

20/24 regions pass (≥ 0.70 recovery)

Sufficiency is the complement to necessity (IIA). Together they establish the evidence subspace as a causal mediator of choice.

Not explained by confounds



Dashed lines = $|\rho| > 0.5$ danger zone

No strong confound detected

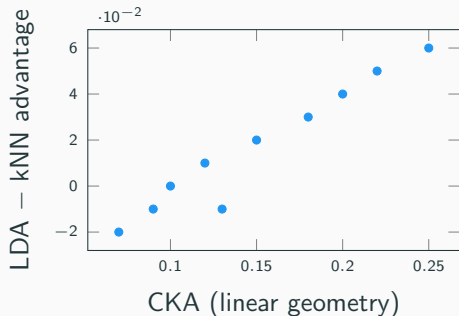
- Neuron count: $\rho = -0.48$ (close but below threshold)
- Firing rate: $\rho = 0.21$ (not significant)
- Trial count: $\rho = -0.21$ (not significant)
- Within-mouse: effect holds per mouse
- Temporal shuffle: IIA $\approx$ shuffle (needs full data)

Pass: exp51 confound control

Novel predictions

Geometric type predicts decoder choice

exp22: Regions where CKA is high (“linear-type”) favor LDA over kNN for decoding; regions where Procrustes is high (“nonlinear-type”) favor kNN.



$$\rho = 0.500, p = 0.00016 (n = 50 \text{ regions})$$

Causal abstraction geometry

exp33: Power-law exponent α correlates with evidence-choice subspace routing across 73 regions.

$$\rho = 0.288, p = 0.014$$

Allen Atlas: null result

exp45: Structural connectivity (Allen Mouse Brain) does not predict functional IIA.

$$\rho = 0.031, p = 0.65$$

This is an honest null. Structural wiring does not predict which regions route evidence causally. Function $\neq$ anatomy at this level.

5 of 7 validity criteria pass cleanly. Two novel predictions work (geometric type, causal abstraction); one is honestly null (Allen Atlas). We report failures alongside successes.

What this means

Summary

Eigenvalue spectra universal; trial content idiosyncratic

Linear $\leftrightarrow$ nonlinear geometry anti-correlated ($\rho = -0.85$)

Choice manifolds topologically flat ($\beta_1 = 0$)

Rotation everywhere (72/73 regions), $10\times$ variation

Geometry recovers anatomy without labels ($k = 3$)

Evidence subspace causally mediates choice (IIA)

Causal effect robust across 4 intervention methods

representational

causal

Limitations and next steps

Honest limitations:

- One dataset (Steinmetz 2019)
- Binary task only
- IIA effect sizes are modest ($\sim 10\%$ flip rate)
- Double dissociation is partial
- Temporal shuffle control is weak
- All analysis is post-hoc (no preregistration)

Next:

- IBL cross-dataset replication
- Allen Atlas anatomical comparison
- Zatka-Haas optogenetic validation (data downloaded, exp47b ready)
- Multi-task generalization (beyond binary choice)

This is exploratory work. Code and results are public. We think geometric validity matters—this is our attempt to take it seriously. Feedback welcome.

Intellectual grounding

This work draws on ideas from three research traditions:

- **Causal representation learning** (Siddharth Mishra-Sharma) — identifiability conditions for when geometric structure in latent spaces reflects true causal variables
- **Relational causal models** (Jensen & Neville) — extending causal inference beyond i.i.d. data to relational and multi-level systems, which is what brain region interactions are
- **Dynamical systems in neuroscience** (Hava Siegelmann) — neural computation as geometry of dynamical trajectories; connections between topology and computational capacity

The validity framework is grounded in philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003).

Code & results: [ZENODO DOI / GITHUB URL]

Neural geometry is structured, causal, and validatable.